3.3.1 Using unitary matrices

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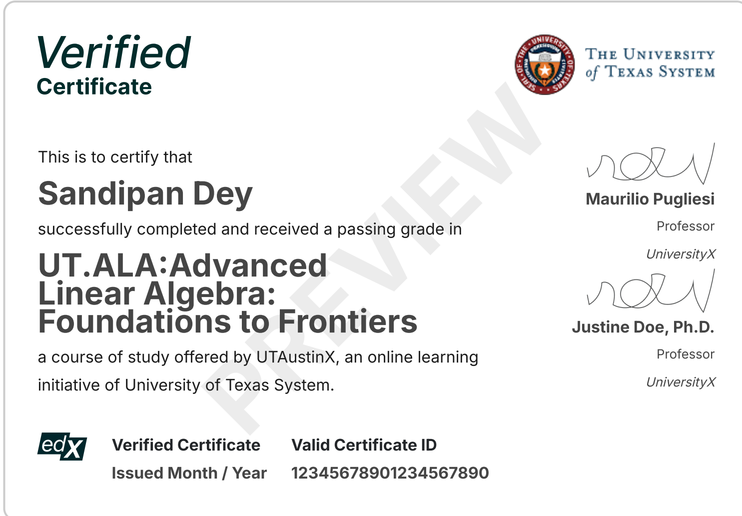

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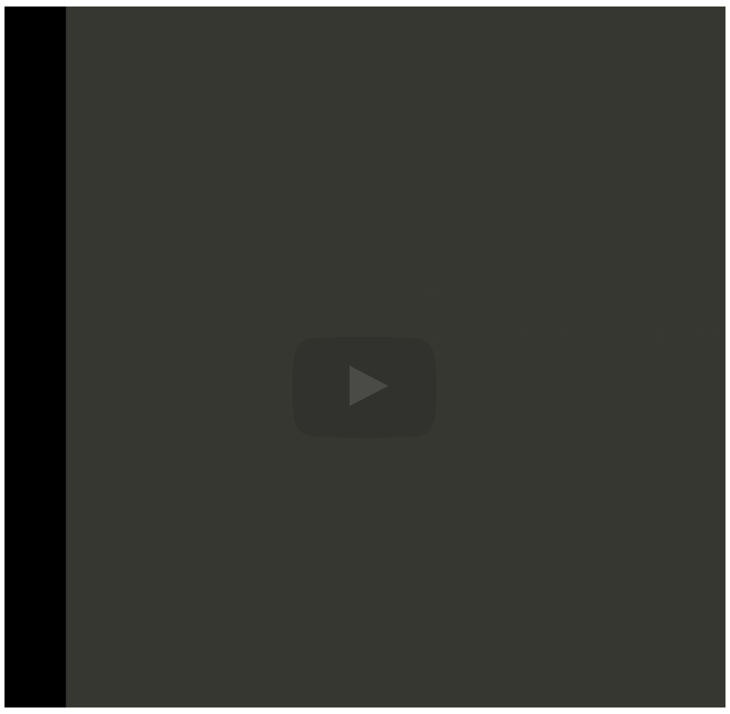

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mutually orthonormal columns, which is the whole purpose of the Gram Schmidt

process. Well, wouldn't it be nice if we could instead compute a sequence of

unitary matrices, and here we use H because we're going to call these

unitary matrices Householder transforms, so we'll have Gauss transforms and

Householder transforms. And wouldn't it be nice if the result of apply this

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Reading assignment

1/1 point (graded)

Read Unit 3.3.1 of the notes. [LINK]

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Homework 3.3.1.1.

1/1 point (graded)

Show that if $A \in \mathbb{C}^{m \times n}$ and $A = Q \begin{pmatrix} R \\ 0 \end{pmatrix}$, where $Q \in \mathbb{C}^{m \times m}$ is unitary and R is upper triangular, then there exists $Q_L \in \mathbb{C}^{m \times n}$ such that $A = Q_L R$, is the QR factorization of A.

☑ done

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Solution:

For a complete solution, see the notes

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